

LPDDR2-S4 Refresh Requirement Parameters

Parameter		Symbol	512Mb	Unit
Number of Banks			4	
Refresh Window $T_{case} \leq 85^{\circ}\text{C}$		tREFW	32	ms
Refresh Window $85^{\circ}\text{C} < T_{case} \leq 105^{\circ}\text{C}$		tREFW	8	ms
Required number of REFRESH commands (min)		R	4096	
Average time between REFRESH commands (for reference only) $T_{case} \leq 85^{\circ}\text{C}$	REFab	tREFI	7.8	us
	REFpb	tREFIpb	Not permitted	
Refresh Cycle time		tRFCab	90	ns
Per Bank Refresh Cycle time		tRFCpb	N/A	ns
Burst Refresh Window = $(4 \times 8 \times tRFCab)$		tREFBW	2.88	us

AC TIMINGS

LPDDR2 AC Timing Table (1,2)

Parameter	Symbol	min max	min t _{CK}	LPDDR2										Unit
				1066	933	800	667	533	466	400	333	266	200	
Max. Frequency		~		533	466	400	333	266	233	200	166	133	100	MHz
Clock Timing														
Average Clock Period	t _{CK} (avg)	min max				2.5	3	3.75	4.3	5	6	7.5	10	ns
Average high pulse width	t _{CH} (avg)	min							0.45					t _{CK} (avg)
		max							0.55					
Average low pulse width	t _{CL} (avg)	min							0.45					t _{CK} (avg)
		max							0.55					
Absolute Clock Period	t _{CK} (abs)	min							t _{CK} (avg)min + t _{JIT} (per),min					ps
Absolute clock HIGH pulse width (with allowed jitter)	t _{CH} (abs), allowed	min							0.43					t _{CK} (avg)
		max							0.57					t _{CK} (avg)
Absolute clock LOW pulse width (with allowed jitter)	t _{CL} (abs), allowed	min							0.43					t _{CK} (avg)
		max							0.57					t _{CK} (avg)
Clock Period Jitter (with allowed jitter)	t _{JIT} (per), allowed	min		-90	-95	-100	-110	-120	-130	-140	-150	-180	-250	ps
		max		90	95	100	110	120	130	140	150	180	250	
Maximum Clock Jitter between two consecutive clock cycles (with allowed jitter)	t _{JIT} (cc), allowed	max		180	190	200	220	240	260	280	300	360	500	ps
Duty cycle Jitter (with allowed jitter)	t _{JIT} (duty), allowed	min							min((t _{CH} (abs),min - t _{CH} (avg),min), (t _{CL} (abs),min - t _{CL} (avg),min)) * t _{CK} (avg)					ps
		max							max((t _{CH} (abs),max - t _{CH} (avg),max), (t _{CL} (abs),max - t _{CL} (avg),max)) * t _{CK} (avg)					ps
Cumulative error across 2 cycles	t _{ERR} (2per), allowed	min		-132	-140	-147	-162	-177	-191	-206	-221	-265	-368	ps
		max		132	140	147	162	177	191	206	221	265	368	
Cumulative error across 3 cycles	t _{ERR} (3per), allowed	min		-157	-166	-175	-192	-210	-227	-245	-262	-314	-437	ps
		max		157	166	175	192	210	227	245	262	314	437	
Cumulative error across 4 cycles	t _{ERR} (4per), allowed	min		-175	-185	-194	-214	-233	-253	-272	-291	-350	-486	ps
		max		175	185	194	214	233	253	272	291	350	486	
Cumulative error across 5 cycles	t _{ERR} (5per), allowed	min		-188	-199	-209	-230	-251	-272	-293	-314	-377	-524	ps
		max		188	199	209	230	251	272	293	314	377	524	
Cumulative error across 6 cycles	t _{ERR} (6per), allowed	min		-200	-211	-222	-244	-266	-288	-311	-333	-399	-555	ps
		max		200	211	222	244	266	288	311	333	399	555	
Cumulative error across 7 cycles	t _{ERR} (7per), allowed	min		-209	-221	-232	-256	-279	-302	-325	-348	-418	-581	ps
		max		209	221	232	256	279	302	325	348	418	581	

LPDDR2 AC Timing Table (1,2)

Parameter	Symbol	min max	min t _{CK}	LPDDR2										Unit
				1066	933	800	667	533	466	400	333	266	200	
Cumulative error across 8 cycles	t _{ERR} (8per), allowed	min		-217	-229	-214	-266	-290	-314	-338	-362	-435	-604	ps
		max		217	229	214	266	290	314	338	362	435	604	
Cumulative error across 9 cycles	t _{ERR} (9per), allowed	min		-224	-237	-249	-274	-299	-324	-349	-374	-449	-624	ps
		max		224	237	249	274	299	324	349	374	449	624	
Cumulative error across 10 cycles	t _{ERR} (10per), allowed	min		-231	-244	-257	-282	-308	-334	-359	-385	-462	-641	ps
		max		231	244	257	282	308	334	359	385	462	641	
Cumulative error across 11 cycles	t _{ERR} (11per), allowed	min		-237	-250	-263	-289	-316	-342	-368	-395	-474	-658	ps
		max		237	250	263	289	316	342	368	395	474	658	
Cumulative error across 12 cycles	t _{ERR} (12per), allowed	min		-242	-256	-269	-296	-323	-350	-377	-403	-484	-672	ps
		max		242	256	269	296	323	350	377	403	484	672	
Cumulative error across n = 13, 14 . . . 49, 50 cycles	t _{ERR} (nper), allowed	min		$t_{ERR}(nper), allowed, min = (1 + 0.68 \ln(n)) * t_{JIT}(per), allowed, min$										ps
		max		$t_{ERR}(nper), allowed, max = (1 + 0.68 \ln(n)) * t_{JIT}(per), allowed, max$										

LPDDR2 AC Timing Table (1,2)

Parameter	Symbol	min max	min t _{CK}	LPDDR2										Unit
				1066	933	800	667	533	466	400	333	266	200	
ZQ Calibration Parameters														
Initialization Calibration Time	t _{ZQINIT}	min		1										us
Long Calibration Time	t _{ZQCL}	min	6	360										ns
Short Calibration Time	t _{ZQCS}	min	6	90										ns
Calibration Reset Time	t _{ZQRESET}	min	3	50										ns
Read Parameters (3)														
DQS output access time from CK/CK#	t _{DQSCK}	min		2500										ps
		max		5500										
DQSCK Delta Short ⁽⁴⁾	t _{DQSCKDS}	max		330	380	450	540	670	770	900	1080	1350	1800	ps
DQSCK Delta Medium ⁽⁵⁾	t _{DQSCKDM}	max		680	780	900	1050	1350	1550	1800	1900	2000	2100	ps
DQSCK Delta Long ⁽⁶⁾	t _{DQSCKDL}	max		920	1050	1200	1400	1800	2100	2400	-	-	-	ps
DQS - DQ skew	t _{DQSQ}	max		200	220	240	280	340	370	400	500	600	700	ps
Data hold skew factor	t _{QHS}	max		230	260	280	340	400	450	480	600	750	1000	ps
DQS Output High Pulse Width	t _{QSH}	min		t _{CH} (abs) - 0.05										t _{CK} (avg)
DQS Output Low Pulse Width	t _{QSL}	min		t _{CL} (abs) - 0.05										t _{CK} (avg)
Data Half Period	t _{QHP}	min		min(t _{QSH} , t _{QSL})										t _{CK} (avg)
DQ / DQS output hold time from DQS	t _{QH}	min		t _{QHP} - t _{QHS}										ps
Read preamble ⁽⁸⁾	t _{RPRE}	min		0.9										t _{CK} (avg)
Read postamble ⁽⁹⁾	t _{RPST}	min		t _{CL} (abs) - 0.05										t _{CK} (avg)
DQS low-Z from clock ⁽⁷⁾	t _{LZ(DQS)}	min		t _{DQSCK} (MIN) - 300										ps
DQ low-Z from clock ⁽⁷⁾	t _{LZ(DQ)}	min		t _{DQSCK} (MIN) - (1.4 * t _{QHS} (MAX))										ps
DQS high-Z from clock ⁽⁷⁾	t _{HZ(DQS)}	max		t _{DQSCK} (MAX) - 100										ps
DQ high-Z from clock ⁽⁷⁾	t _{HZ(DQ)}	max		t _{DQSCK} (MAX) + (1.4 * t _{DQSQ} (MAX))										ps

LPDDR2 AC Timing Table (1,2)

Parameter	Symbol	min max	min t _{CK}	LPDDR2										Unit	
				1066	933	800	667	533	466	400	333	266	200		
Write Parameters (3)															
DQ and DM input hold time (V _{ref} based)	t _{DH}	min		210	235	270	350	430	450	480	600	750	1000	ps	
DQ and DM input setup time (V _{ref} based)	t _{DS}	min		210	235	270	350	430	450	480	600	750	1000	ps	
DQ and DM input pulse width	t _{DIPW}	min		0.35										t _{CK} (avg)	
Write command to 1st DQS latching transition	t _{DQSS}	min		0.75										t _{CK} (avg)	
		max		1.25											
DQS input high-level width	t _{DQSH}	min		0.4										t _{CK} (avg)	
DQS input low-level width	t _{DQSL}	min		0.4										t _{CK} (avg)	
DQS falling edge to CK setup time	t _{DSS}	min		0.2										t _{CK} (avg)	
DQS falling edge hold time from CK	t _{DSH}	min		0.2										t _{CK} (avg)	
Write postamble	t _{WPST}	min		0.4										t _{CK} (avg)	
Write preamble	t _{WPPE}	min		0.35										t _{CK} (avg)	

LPDDR2 AC Timing Table (1,2)

Parameter	Symbol	min max	min t _{CK}	LPDDR2										Unit	
				1066	933	800	667	533	466	400	333	266	200		
CKE Input Parameters															
CKE min. pulse width (high and low pulse width)	t _{CKE}	min	3	3										t _{CK} (avg)	
CKE input setup time	t _{ISCKE} (10)	min		0.25										t _{CK} (avg)	
CKE input hold time	t _{IHCKE} (11)	min		0.25										t _{CK} (avg)	
Command Address Input Parameters (3)															
Address and control input setup time (V _{ref} based)	t _{IS} (12)	min		220	250	290	370	460	520	600	740	900	1150	ps	
Address and control input hold time (V _{ref} based)	t _{IH} (12)	min		220	250	290	370	460	520	600	740	900	1150	ps	
Address and control input pulse width	t _{IPW}	min		0.40										t _{CK} (avg)	
Boot Parameters (10 MHz - 55 MHz) (13,14,15)															
Clock Cycle Time	t _{CKb}	max	-	100										ns	
		min	81												
CKE Input Setup Time	t _{ISCKEb}	min	-	2.5										ns	
CKE Input Hold Time	t _{IHCKEb}	min	-	2.5										ns	
Address & Control Input Setup Time	t _{ISb}	min	-	1150										ps	
Address & Control Input Hold Time	t _{IHb}	min	-	1150										ps	
DQS Output Data Access Time from CK/CK#	t _{DQSCkCb}	min	-	2.0										ns	
		max	10.0												
Data Strobe Edge to Ouput Data Edge t _{DQSQb} - 1.2	t _{DQSQb}	max	-	1.2										ns	
Data Hold Skew Factor	t _{QHSb}	max	-	1.2										ns	
Mode Register Parameters															
MODE REGISTER Write command period	t _{MRW}	min	5	5										t _{CK} (avg)	
Mode Register Read command period	t _{MRR}	min	2	2										t _{CK} (avg)	

LPDDR2 AC Timing Table ^(1,2)

Parameter	Symbol	min	max	min t _{CK}	LPDDR2 SDRAM Core Parameters (16)										LPDDR2					Unit
					1066	933	800	667	533	466	400	333	266	200						
Read Latency	RL	min		3	8	7	6	5	4		3	3					t _{CK} (avg)			
Write Latency	WL	min		1	4	4	3	2	2		1	1					t _{CK} (avg)			
ACTIVE to ACTIVE command period	t _{RC}	min			t _{RAS} + t _{RPab} (with all-bank Precharge) t _{RAS} + t _{RPpb} (with per-bank Precharge)													ns		
CKE min. pulse width during Self-Refresh (low pulse width during Self-Refresh)	t _{CKESR}	min		3				15			15						ns			
Self refresh exit to next valid command delay	t _{XSR}	min		2	t _{RRCab} + 10													ns		
Exit power down to next valid command delay	t _{XP}	min		2				7.5			7.5						ns			
LPDDR2-S4 CAS to CAS delay	t _{CCD}	min		2				2			2						t _{CK} (avg)			
Internal Read to Precharge command delay	t _{RTP}	min		2				7.5			7.5						ns			
RAS to CAS Delay ⁽¹⁹⁾	t _{RCD}	Fast (min)		3				15			15						ns			
		Typ (min)		3				18			18						ns			
Row Precharge Time ⁽¹⁹⁾ (single bank)	t _{RPpb}	Fast (min)		3				15			15						ns			
		Typ (min)		3				18			18						ns			
Row Precharge Time ⁽¹⁹⁾ (all banks)	t _{RPab} 4-bank	Fast (min)		3				15			15						ns			
		Typ (min)		3				18			18						ns			
Row Active Time	t _{RAS}	min		3				42			42						ns			
		max		-				70			70						us			
Write Recovery Time	t _{WR}	min		3				15			15						ns			
Internal Write to Read Command Delay	t _{WTR}	min		2				7.5			10						ns			
Active bank A to Active bank B	t _{RBD}	min		2				10			10						ns			
Four Bank Activate Window	t _{FAW}	min		8				50			50	60					ns			
Minimum Deep Power Down Time	t _{DPD}	min						500			500						us			

Table 103 — LPDDR2 AC Timing Table

Parameter	Symbol	min max	min t _{CK}	LPDDR2								Unit	
				1066	933	800	667	533	466	400	333		266
LPDDR2 Temperature De-Rating ⁽¹⁷⁾													
Core Timings Temperature De-Rating	t _{DQCK} (Derated)	max		5620						6000			ps
	t _{RCD} (Derated)	min							t _{RCD} + 1.875				ns
	t _{RC} (Derated)	min							t _{RC} + 1.875				ns
	t _{RAS} (Derated)	min							t _{RAS} + 1.875				ns
	t _{RP} (Derated)	min							t _{RP} + 1.875				ns
	t _{RRD} (Derated)	min							t _{RRD} + 1.875				ns

